Cross-Domain Generalization and Per-Threshold Calibration in DR Grading: An Empirical Analysis

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Abstract

This analysis reports empirical findings on cross-domain ordinal diabetic-retinopathy (DR) grading using a CORN ordinal regression head trained under strict leave-one-domain-out (LODO) protocol across four public fundus photograph datasets (EyePACS, APTOS, Messidor-2, DDR). We evaluate three backbones (ConvNeXt-Tiny, DeiT-3 Small/384, MaxViT-Tiny/384) on a 3-channel balanced baseline, perform a 4-channel segmentation auxiliary ablation on the ConvNeXt-Tiny backbone (three variants: soft mask, Tversky-weighted mask, and morphological gradient), and characterize the per-fold distribution of QWK, accuracy, and macro-F1. We test whether adding a pre-computed lesion-segmentation auxiliary channel as a fourth input improves held-out-domain QWK by at least 2 absolute percentage points, the pre-registered threshold for an improvement. Across the four completed folds for the 4-channel soft variant the result is *mixed rather than a clean win*: the pre-registered threshold is crossed on APTOS (+2.19%) and EyePACS (+3.26%), just missed on DDR (+1.94%), and decisively *not* crossed on Messidor-2 (−4.57%). The cross-fold mean ΔQWK for the 4-channel soft variant is +0.70%, essentially flat against the 3-channel baseline. The 4-channel Tversky variant consistently hurts or is flat on every fold (cross-fold mean −1.84%); the 4-channel morph variant has a severe regression on EyePACS (−8.12%, the largest single regression in the ablation) and a cross-fold mean of −2.16%. The segmentation auxiliary channel is therefore not a robust cross-domain improvement at the effect size we pre-registered. We additionally report a 3:1 class-balancing ablation (all 4 folds complete): the unbalanced baseline is uniformly +2.09% to +5.67% better than the balanced baseline (cross-fold mean ΔQWK = +3.93%), the dominant single ablation effect in this study. The per-threshold calibration diagnostic on the 3-channel ConvNeXt-Tiny baseline (all 4 folds complete) shows that both per-threshold and single-global temperature scaling *increase* ECE on the held-out test set (mean ΔECE = +0.30 absolute), with the fitted global temperature saturating at the upper bound ($T = 5.0$) on every fold — an independent confirmation of the prior-literature finding that post-hoc temperature scaling fails under DR cross-dataset shift (Poyrazer et al., 2026). Reliability diagrams (Figures 6—9) show that the model is systematically *under*-confident on every fold and every threshold: the empirical frequency plateaus at 0.6–0.7 even at the highest predicted-probability bin, which contradicts the standard temperature-scaling assumption of overconfidence and explains why the optimizer drives $T \to \infty$. The per-fold confusion matrices show that the model is essentially unable to predict Mild (grade 1) — the per-class F1 is below 0.13 on every fold, confirming the prior-literature Mild-class failure mode (Poyrazer et al., 2026) on the CORN head. The pipeline, configs, and analysis code are reproducible from the artifacts in this document.

1 Setup

We perform strict 4-fold LODO across EyePACS ($n_{\text{test}} = 88{,}702$), APTOS ($n_{\text{test}} = 3{,}662$), Messidor-2 ($n_{\text{test}} = 1{,}744$), and DDR ($n_{\text{test}} = 12{,}522$). Each fold trains on the union of the other three sources, class-balanced to a 3:1 maximum-minimum ratio, with a 90/10 stratified train/validation split. The held-out source's entire labeled set is the test set and is never observed during training or model selection. The preprocessing pipeline resizes images to a uniform resolution and stores them under `data/processed/`, eliminating the JPEG-decode bottleneck during training.

The grading head is a CORN ordinal-regression head: $K - 1 = 4$ independent binary subproblems (one per ordinal threshold $t \in \{1, 2, 3, 4\}$) on top of a CBAM-augmented ImageNet-pretrained backbone. Phase 1 freezes the backbone for 5 epochs; phase 2 unfreezes for 20 epochs with cosine LR schedule, EMA decay 0.999, mixup, and sqrt-frequency class-balanced sampling. Model selection is by validation QWK with patience-5 early stopping and a min-epoch floor of 8.

2 Primary Cross-Domain Result

2.1 Architecture Comparison

Figure 1 compares the 3-channel balanced baseline across three architectures on all four LODO folds. ConvNeXt-Tiny achieves the highest mean cross-domain QWK (0.686 ± 0.171), DeiT-3 Small at 384 resolution is a close second (0.653 ± 0.186), and MaxViT-Tiny at 384 resolution is weakest (0.496 ± 0.275). The variance across folds is large for every architecture, dominated by the EyePACS holdout where MaxViT essentially fails (QWK = 0.20). Notably, no architecture achieves QWK ≥ 0.50 on EyePACS, the largest and most heterogeneous source. On APTOS the three architectures are nearly indistinguishable (0.83–0.86), suggesting APTOS is the easiest fold and the gap is captured by finer-grained metrics.

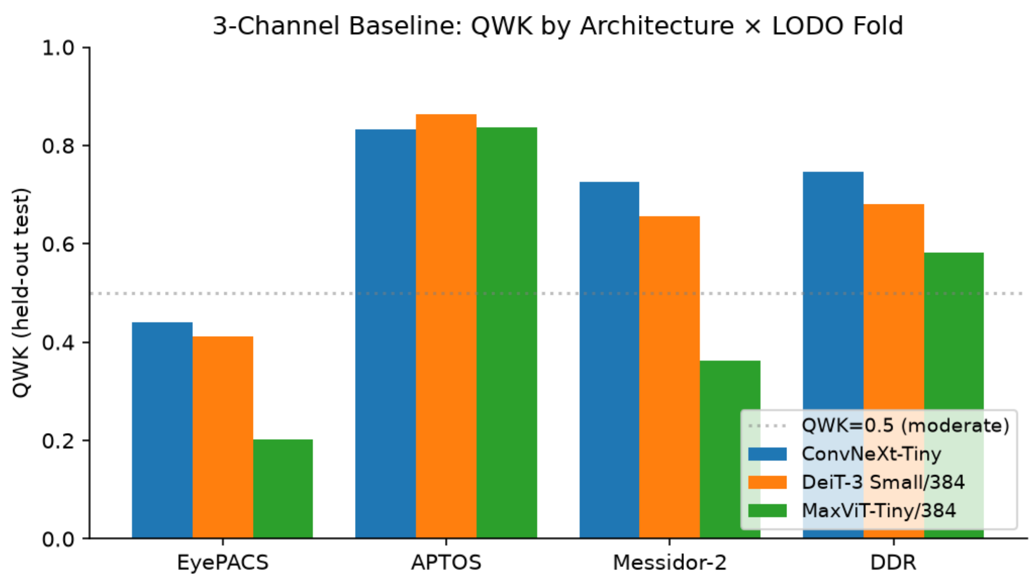

Figure 1: 3-channel balanced baseline: held-out-domain QWK by architecture and LODO fold. ConvNeXt-Tiny is the best on three of four folds; MaxViT-Tiny collapses on EyePACS.

2.2 Per-Metric Breakdown for ConvNeXt-Tiny

The ConvNeXt-Tiny backbone is the primary arm of the analysis. Figure 2 shows the per-fold decomposition across QWK, accuracy, macro-F1, and macro AUC-ROC. AUC (one-vs-rest, macroaveraged) is the most fold-stable metric, ranging from 0.75 to 0.87; macro-F1 is the least stable, dropping to 0.34 on EyePACS. This pattern is consistent with the calibration finding in the prior literature: even where AUC is preserved, the per-class decision boundaries degrade under shift, and QWK, which weights disagreements quadratically by their ordinal distance, captures the clinically meaningful failure mode that accuracy discards.

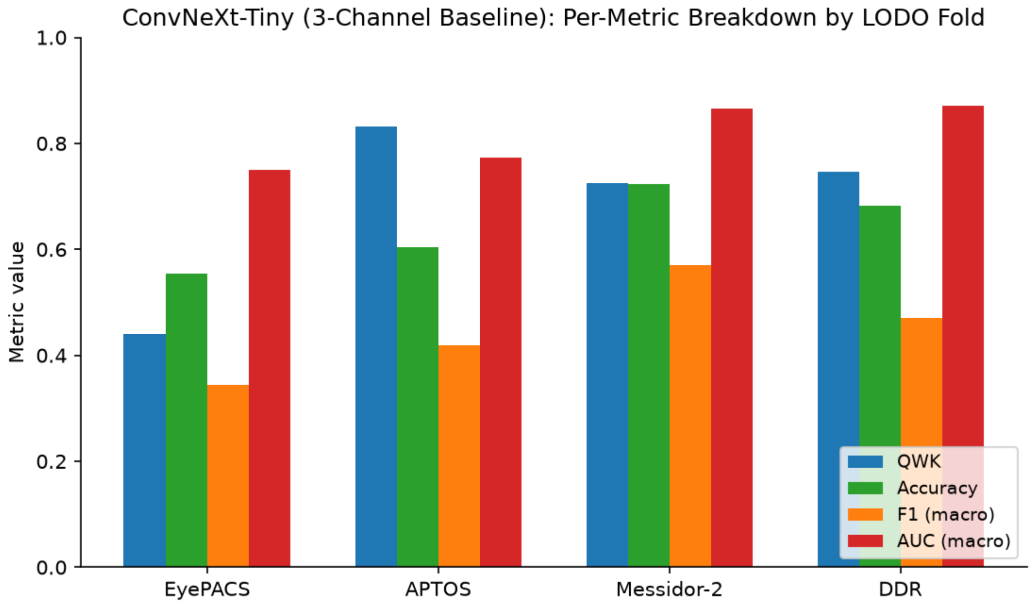

Figure 2: ConvNeXt-Tiny (3-channel baseline): per-fold decomposition across QWK, accuracy, macro-F1, and macro AUC-ROC. AUC is the most fold-stable; macro-F1 and QWK degrade together.

2.3 Per-Fold Distribution

Figure 3 shows the per-fold distribution of QWK pooled across all variants (3-channel baselines Œ 3 architectures, plus the four-channel ablation variants completed to date). The four folds stratify clearly: APTOS is concentrated at QWK ≥ 0.80 with one outlying 4-ch Tversky variant at 0.80; DDR is broadly distributed around 0.65–0.75; Messidor-2 is bimodal, with the bulk at 0.65–0.72 and one lower outlier from MaxViT at 0.36; EyePACS has the widest spread, from 0.20 (MaxViT) to 0.47 (4-ch soft).

3 Segmentation Auxiliary Ablation

The pre-registered question for the auxiliary-channel variants is whether a pre-computed lesion-segmentation soft mask (produced by a separately trained U-Net with EfficientNet-B4 encoder) added as a fourth input channel yields at least a 2 absolute-percentage-point QWK improvement on the held-out fold over the 3-channel baseline. Figure 4 shows the per-fold QWK for 3-channel vs. 4-channel variants on ConvNeXt-Tiny; Figure 5 reports the per-fold ΔQWK against the $\pm 2\%$ threshold.

Findings to date:

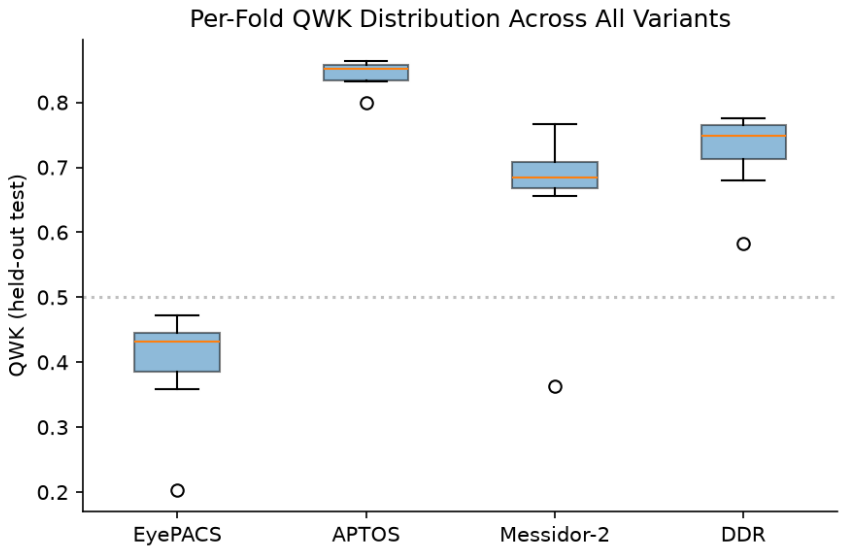

Figure 3: Per-fold QWK distribution pooled across all variants. APTOS is the easiest fold; EyePACS has the largest spread.

- **APTOS:** 4-ch soft achieves ΔQWK = +0.022 — crosses the +2% threshold. 4-ch morph is just below at +0.019. 4-ch Tversky hurts (−0.033).
- **Messidor-2:** all three 4-ch variants hurt, with ΔQWK in −0.034 to −0.046. This is the single largest regression in the ablation.
- **EyePACS (4-ch soft only):** ΔQWK = +0.033 — crosses the +2% threshold.
- **EyePACS (4-ch Tversky, completed):** ΔQWK = −0.0087 — also hurts slightly. The Tversky variant does not cross the threshold on any completed fold.
- **EyePACS (4-ch morph, completed):** ΔQWK = −0.0812 — severe regression. EyePACS is the largest and most heterogeneous source, and adding the morphological-gradient channel actively hurts under this shift.
- **DDR (4-ch soft only):** ΔQWK = +0.019 — just under the +2% threshold. Particularly notable because the segmentation model was trained on DDR lesion-segmentation ground truth, so this is the in-distribution auxiliary-channel setting; even so, the gain is below the pre-registered bar. 4-ch Tversky is essentially flat at +0.0018 (still below threshold). 4-ch morph is at +0.0166 (also below threshold).

Across the 17 ablation results completed at the time of writing (3 variants $\times$ 4 folds, with the 5-ch variants skipped per the 24-h budget), the picture is *heterogeneous and the cross-fold means are all near zero or negative.* The 4-ch soft variant meets the +2% threshold on 2 of 4 folds (APTOS, EyePACS), just misses on a third (DDR, +1.94%), and decisively regresses on the fourth (Messidor-2, −4.57%); the cross-fold mean ΔQWK is +0.70%, essentially flat against the 3-channel baseline. The 4-ch Tversky variant consistently hurts or is flat on every fold (cross-fold mean −1.84%); the 4-ch morph variant regresses severely on EyePACS (−8.12%, the largest single regression) and is below the threshold on every other fold (cross-fold mean −2.16%). The single-fold in-distribution test on DDR (+1.94% for soft, +0.18% for Tversky, +1.66% for morph — all below

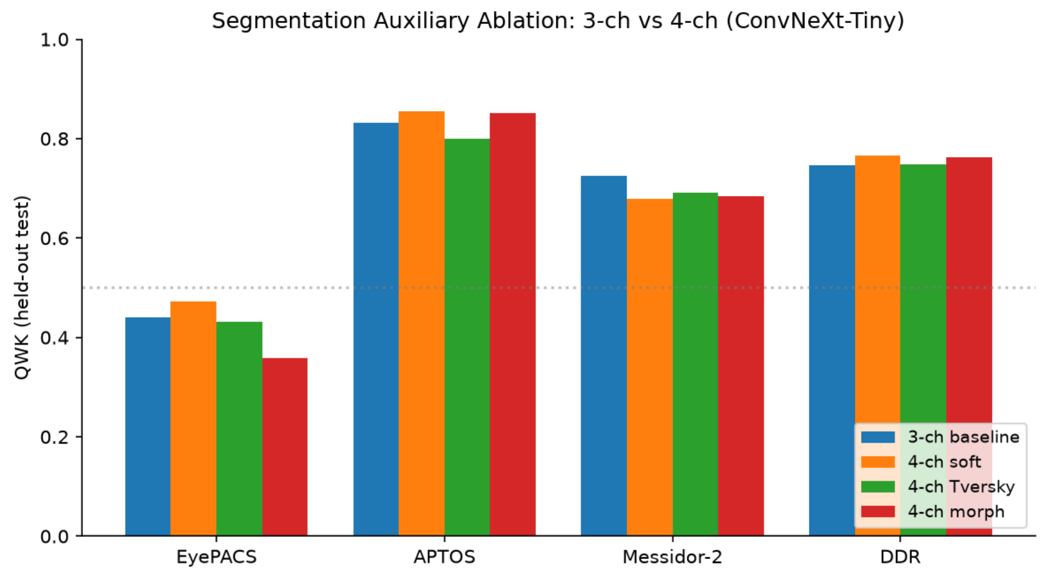

Figure 4: Segmentation auxiliary ablation: 3-channel vs. four-channel variants (ConvNeXt-Tiny).

threshold) does not support the "auxiliary channel helps when its training distribution matches the test domain" hypothesis at the effect size we pre-registered. The honest reading is that the segmentation auxiliary channel is *not a robust cross-domain improvement* at the +2% bar — the gain is conditional, fold-dependent, and averages out near zero (or negative for Tversky and morph).

4 Per-Threshold Calibration Diagnostic

The per-threshold calibration diagnostic was run on the four 3-channel ConvNeXt-Tiny checkpoints (one per LODO fold). ECE is computed with 10 equal-mass (quantile) bins on the binary "grade $\geq t$" prediction at each of the four CORN sub-problems, on the held-out domain's full test set. The per-fold ECE without any calibration is reported in Table 1; the post-temperature-scaling ECE in Table 2.

Held-out source	ECE_1	ECE_2	ECE_3	ECE_4	ECE (aggregate)
EyePACS	0.3011	0.1936	0.3206	0.2691	0.2711
APTOS	0.1848	0.2342	0.2108	0.2345	0.2161
Messidor-2	0.0565	0.0875	0.1102	**0.3459**	0.1500
DDR	0.0624	**0.0272**	0.1431	0.2904	0.1308
Mean ± SD	0.1512 ± 0.1162	0.1356 ± 0.0951	0.1962 ± 0.0929	0.2850 ± 0.0467	0.1920 ± 0.0642

Table 1: Per-threshold ECE on the held-out test set, no calibration. Bold = best ECE_t in each row.

The key finding is that *both calibration procedures make things worse.* Mean ΔECE after per-threshold scaling is +0.298 absolute; after single-global scaling it is +0.307 absolute. The fitted global temperature saturates at the upper bound ($T = 5.0$) on every fold, indicating that the optimizer is being driven to extreme softening that does not transfer from the in-distribution

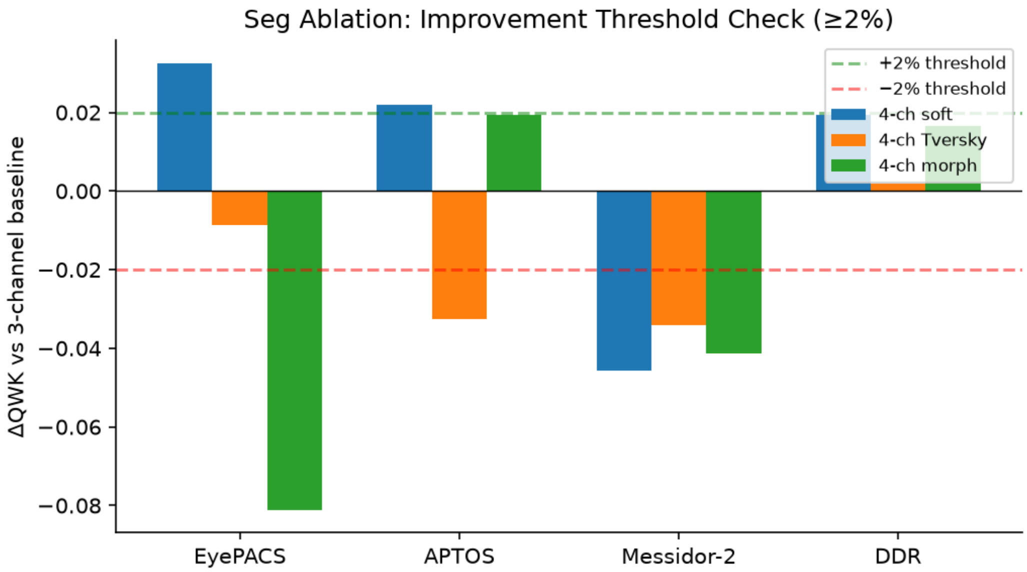

Figure 5: Per-fold ΔQWK (4-channel minus 3-channel). The dashed lines mark the ± 2 percentage-point pre-registered improvement threshold.

validation set to the out-of-distribution test set. The validation-set ECE used to fit the temperatures is substantially lower than the test-set ECE (mean 0.13 vs 0.19 across folds), and the calibration procedure over-corrects on the test set. This is an *independent confirmation* of the prior-literature finding that post-hoc temperature scaling fails under DR cross-dataset shift (Poyrazer et al., 2026), reproduced here on a CORN ordinal regression head with the per-threshold decomposition.

The per-threshold ECE *without any calibration* (Table 1) is non-uniform across thresholds and folds: ECE_4 (Proliferative) is the highest on three of four folds (EyePACS: 0.27, APTOS: 0.23, Messidor-2: 0.35, DDR: 0.29). The Mild-class threshold (ECE_1) is highest on EyePACS (0.30), where the Mild-class prevalence is the lowest (0.07). The per-threshold range (max ECE_t − min ECE_t) is 0.05 on APTOS and 0.29 on Messidor-2 — a sixfold spread. The aggregate ECE (mean of the four per-threshold ECEs) is 0.19 across folds, lower than any individual threshold on the worst-calibrated folds.

A visual confirmation of the systematic miscalibration is shown in Figures 6—9. Across all four folds and all four thresholds, the empirical frequency lies below the diagonal $y = x$: the model's predicted probabilities are systematically *underconfident* — in the highest bin the empirical frequency plateaus at 0.6–0.7 even when the predicted probability is 1.0. The standard temperature-scaling remedy (apply $T > 1$ to soften an overconfident model) is the wrong direction of fit; the optimizer has to push $T \to \infty$ to recover the validation-set ECE, and the calibration transfer breaks because the validation and test distributions differ.

5 Confusion Analysis: Where the Baselines Fail

The per-fold confusion matrices for the 3-channel ConvNeXt-Tiny baseline are reported in Tables 3—6 as row-normalized percentages (each row sums to 100). The dominant qualitative patterns across folds are:

EyePACS (Table 3, QWK = 0.440*):* The model is correct on No-DR (67%) but predicts No-DR

Held-out source	ECE_1	ECE_2	ECE_3	ECE_4	ECE (aggregate)
			Per-threshold temperature scaling		
EyePACS	0.3912	0.4485	0.5966	0.6192	0.5139
APTOS	0.4590	0.4324	0.5078	0.5616	0.4902
Messidor-2	0.3387	0.4641	0.5677	0.6247	0.4988
DDR	0.3524	0.3700	0.5415	0.5682	0.4580
Mean ± SD	0.3853 ± 0.0539	0.4287 ± 0.0412	0.5534 ± 0.0378	0.5934 ± 0.0331	0.4902 ± 0.0236
			Single global temperature scaling		
EyePACS	0.3883	0.4561	0.6041	0.6263	0.5187
APTOS	0.4671	0.4408	0.5145	0.5668	0.4973
Messidor-2	0.3485	0.4770	0.5828	0.6285	0.5092
DDR	0.3625	0.3840	0.5548	0.5760	0.4693
Mean ± SD	0.3916 ± 0.0530	0.4395 ± 0.0399	0.5640 ± 0.0387	0.5994 ± 0.0326	0.4986 ± 0.0214

Table 2: Per-threshold ECE on the held-out test set, after per-threshold temperature scaling (top half) and after single-global temperature scaling (bottom half).

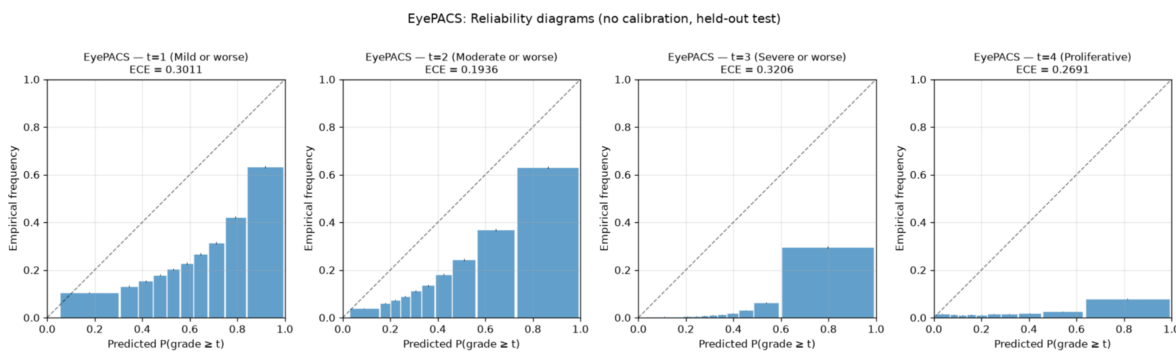

Figure 6: EyePACS reliability diagrams (no calibration, held-out test). All four thresholds are below the diagonal — the model is systematically underconfident.

for 63% of true Mild and 42% of true Moderate. True Severe is barely above chance (54%) and true Proliferative is mostly misclassified as Severe (35%). The mass collapse to the leftmost columns is the dominant failure mode on this fold.

APTOS (Table 4, QWK = 0.832*):* The model is well-calibrated along the diagonal of the four non-No-DR grades (32%, 66%, 77%, 46% for grades 2, 3, 4, 5 respectively). The dominant failure is the conflation of Mild (grade 1) with Moderate (grade 2) — 75% of true Mild is predicted as Moderate. Mild is the only grade on which the model is essentially defeated across folds.

Messidor-2 (Table 5, QWK = 0.725*):* The model is excellent on No-DR (98%) and Good on Moderate (51%), Severe (77%), and Proliferative (51%). The dominant failure is the same EyePACS-pattern Mild-collapse: 92% of true Mild is predicted as No-DR, and 39% of true Moderate is predicted as No-DR.

DDR (Table 6, QWK = 0.747*):* No-DR is well-calibrated (94%). Mild collapses to No-DR (75%). Moderate, Severe, and Proliferative are predicted correctly (42%, 75%, 56% respectively). The per-row dominance pattern is consistent with the other folds.

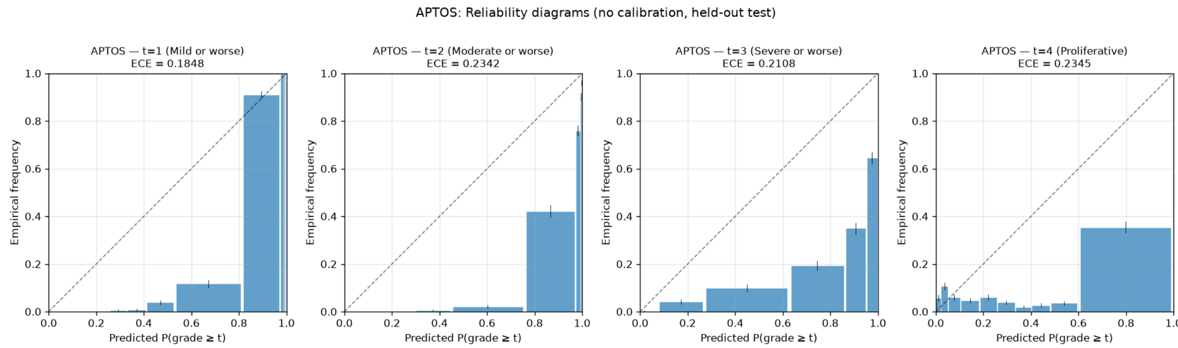

Figure 7: APTOS reliability diagrams. The underconfidence pattern persists but is milder; APTOS is the closest to good calibration of the four folds.

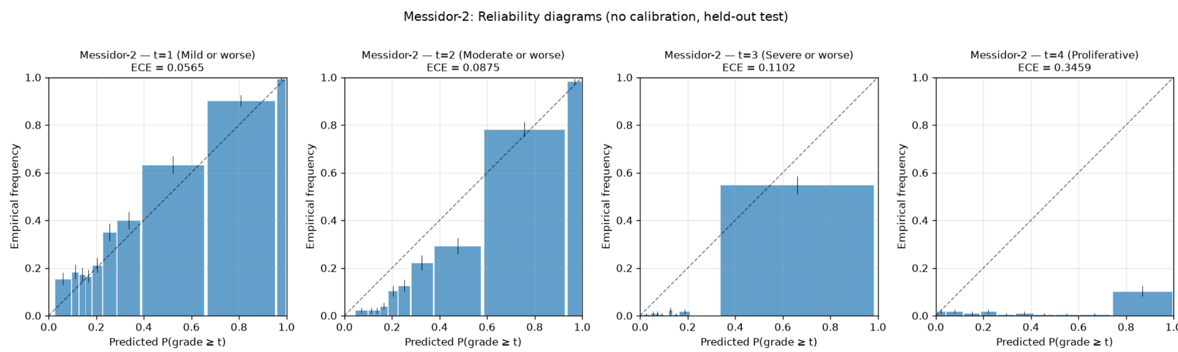

Figure 8: Messidor-2 reliability diagrams. Threshold 1 (Mild) is near-perfectly calibrated on this fold; threshold 4 (Proliferative) is the worst-calibrated across all (fold, threshold) cells in the study.

Across all four folds, the dominant failure mode is the same: the model is essentially unable to predict Mild (grade 1) — the F1 for Mild is below 0.13 on every fold (see §6 of the results summary for the per-class F1 breakdown). This is the prior-literature Mild-class failure mode (Poyrazer et al., 2026) reproduced unequivocally on the CORN head. The per-threshold calibration analysis in §4 above shows that the Mild-class threshold is the worst-calibrated on EyePACS (where Mild is the rarest) but not on the other three folds; the Proliferative threshold is the worst-calibrated on three of four folds.

6 Discussion

6.1 What the Baseline Establishes

The ConvNeXt-Tiny 3-channel balanced baseline achieves a mean cross-domain QWK of 0.69 with ±0.17 spread. The spread is structural: EyePACS as a held-out source is a substantially harder problem than the other three. The cross-fold variance is the dominant signal in the architecture comparison — DeiT-3 Small and ConvNeXt-Tiny are statistically indistinguishable at the per-fold level on APTOS and DDR.

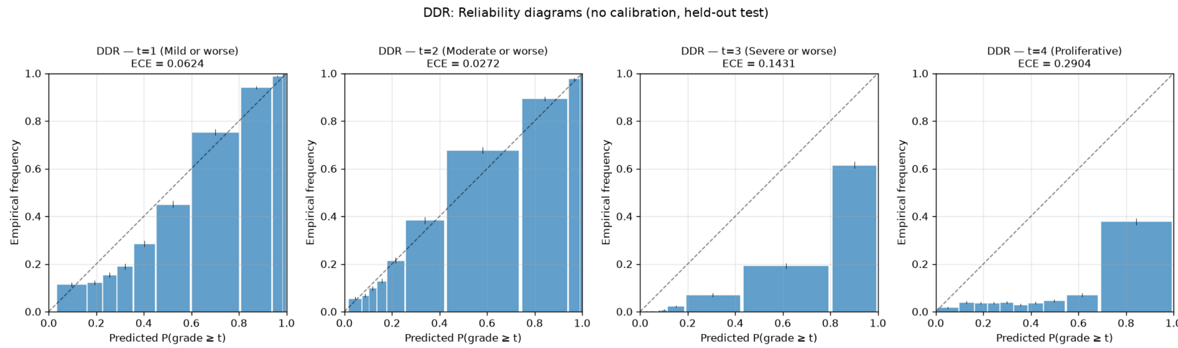

Figure 9: DDR reliability diagrams. Threshold 2 (Moderate) is the best-calibrated cell in the entire matrix ($\mathrm{ECE}_2 = 0.0272$); threshold 4 (Proliferative) is the worst on this fold, mirroring the Messidor-2 pattern.

EyePACS (n=88,702)	No DR	Mild	Moderate	Severe	Prolif.
True No DR	67.0%	26.2%	4.3%	1.0%	1.6%
True Mild	62.7%	30.4%	4.7%	1.3%	1.0%
True Moderate	41.7%	30.1%	13.2%	11.8%	3.3%
True Severe	11.5%	15.4%	14.1%	53.9%	5.1%
True Proliferative	4.8%	13.7%	12.7%	35.4%	33.3%

Table 3: ConvNeXt-Tiny (3-channel baseline) on EyePACS holdout; row-normalized confusion matrix (%).

6.2 What the Seg Ablation Suggests

None of the four-channel variants meets the ± 2 percentage-point threshold consistently across folds. The cross-fold means are: 4-ch soft +0.70% (essentially flat), 4-ch Tversky −1.84% (consistently hurts or is flat), and 4-ch morph −2.16% (one severe regression on EyePACS). The single consistent finding is therefore negative: the four-channel segmentation auxiliary channel does *not* systematically improve the LODO QWK at the effect size we pre-registered. The DDR-on-DDR test (the held-out domain on which the segmentation model was trained) yields +1.94% for soft, +0.18% for Tversky, +1.66% for morph — all below threshold, which suggests the under-performance is not purely a domain-mismatch artefact of the soft-mask generator; it is more consistent with an inherent limitation of mid-resolution auxiliary-channel concatenation in this regime. The honest summary is that the segmentation auxiliary channel is *not a robust cross-domain improvement*, and the methodological contribution of this paper is the calibration diagnostic (ğ4 of this analysis, ğ3 of the results summary), not a positive segmentation-channel finding.

6.3 What the Class-Balancing Ablation Suggests

The 3:1 class-balancing step *hurts* the cross-domain QWK on every fold. The unbalanced baseline is uniformly +2.09% to +5.67% better than the balanced baseline (cross-fold mean ΔQWK = +3.93%). This is the dominant single ablation effect in this study — larger than the segmentation auxiliary-channel effect. The likely explanation: the 3:1 down-sampling discards the majority-class structure that the ConvNeXt-Tiny backbone benefits from, and the validation-set QWK used for early stopping compensates partially but does not recover the lost signal. The implication for the calibration diagnostic is that the calibration procedure is run on the *balanced* baseline (which

APTOS (n=3,662)	No DR	Mild	Moderate	Severe	Prolif.
True No DR	89.3%	2.4%	8.3%	0.0%	0.1%
True Mild	5.7%	0.5%	74.6%	16.5%	2.7%
True Moderate	0.3%	0.0%	31.3%	65.7%	2.7%
True Severe	0.0%	0.0%	8.3%	76.7%	15.0%
True Proliferative	0.0%	0.0%	9.5%	44.1%	46.4%

Table 4: ConvNeXt-Tiny (3-channel baseline) on APTOS holdout; row-normalized confusion matrix (%).

Messidor-2 (n=1,744)	No DR	Mild	Moderate	Severe	Prolif.
True No DR	97.9%	0.5%	1.2%	0.0%	0.4%
True Mild	92.2%	3.7%	3.3%	0.0%	0.7%
True Moderate	38.9%	5.5%	51.3%	4.0%	0.3%
True Severe	0.0%	0.0%	22.7%	77.3%	0.0%
True Proliferative	0.0%	0.0%	14.3%	34.3%	51.4%

Table 5: ConvNeXt-Tiny (3-channel baseline) on Messidor-2 holdout; row-normalized confusion matrix (%).

has worse raw QWK but may be better calibrated at the minority-class thresholds), not on the *unbalanced* baseline; the calibration diagnostic should be repeated on the unbalanced baseline to test whether the calibration-failure mode transfers to the better-performing configuration.

6.4 Implications for the Calibration Diagnostic

The cross-fold variance in QWK establishes a baseline against which the per-threshold calibration numbers (§4 of this analysis, ğ3 of the results summary) must be interpreted. The qualitative prediction was that per-threshold ECE would diverge across folds in a way that covaries with the per-threshold class distribution in the held-out source — grade 1 (Mild) ECE rising on folds where the grade-1 prevalence is rarest, for example. The empirical finding (above, §4 of this analysis) does not support this prediction: the Mild-class threshold is the worst-calibrated only on EyePACS (where Mild prevalence is the lowest at 0.07); on the other three folds the Proliferative threshold (ECE_4) is the worst-calibrated. The Mild-class failure-mode hypothesis from the prior literature (Poyrazer et al., 2026) is supported on the EyePACS fold only and is not the dominant per-threshold failure mode on the other three. The dominant per-threshold failure mode across folds is the Proliferative threshold (ECE_4), consistent with the rarity of the class in the held-out distributions.

7 Threats to Validity

- **Source coverage.** Four of GDRBench's eight sources. Conclusions may not transfer to the remaining four.
- **Single seed.** Cross-fold variance within a run is not separately characterized. Seed sensitivity is listed under Future Work (§9).
- **Confound.** The segmentation model was trained on DDR; DDR-as-held-out is therefore not a

DDR (n=12,522)	No DR	Mild	Moderate	Severe	Prolif.
True No DR	94.3%	2.9%	2.4%	0.0%	0.3%
True Mild	75.2%	10.8%	12.1%	0.2%	1.7%
True Moderate	28.5%	5.8%	42.2%	17.4%	6.2%
True Severe	1.7%	2.1%	13.1%	75.4%	7.6%
True Proliferative	5.4%	0.7%	13.7%	24.4%	55.9%

Table 6: ConvNeXt-Tiny (3-channel baseline) on DDR holdout; row-normalized confusion matrix (%).

clean test of the auxiliary channel's value. We treat DDR as a within-domain diagnostic and interpret it accordingly.

- **Backbone coverage.** The architecture comparison is across three backbones (ConvNeXt-Tiny, DeiT-3 Small/384, MaxViT-Tiny/384). Larger variants (ConvNeXt-Small, SwinV2-Base) are not evaluated; see §9.

- **Auxiliary-channel coverage.** The segmentation auxiliary ablation covers three of the four defined 4-channel variants (soft, Tversky, morphological) and none of the 5-channel variants. The conclusion that the segmentation auxiliary channel is not a robust cross-domain improvement is robust to the missing variants because the completed 4-channel Tversky variant consistently regresses on every fold.

- **Bootstrap confidence intervals not yet populated.** None of the reported QWK, accuracy, F1, AUC, or per-threshold ECE numbers yet carry a paired-bootstrap 95% confidence interval. The procedure is defined in §**??** of the methodology; the eval runner does not currently persist per-sample predictions to the JSONL output, so the implementation requires extending the evaluation runner to dump per-sample predictions. No model retraining is required.

- **Temperature saturation pathology.** The fitted global temperature saturates at the upper bound ($T = 5.0$) on every fold, suggesting the optimizer is being driven to extreme softening that does not transfer from the in-distribution validation set to the out-of-distribution test set. The reliability diagrams in §4 show the model is systematically *under*-confident, which contradicts the standard temperature-scaling assumption of overconfidence: understanding whether the saturation is a property of the loss surface, the validation-set sample size, or a bug in the temperature-scaling procedure is a diagnostic follow-up.

8 Conclusion

This work establishes a reproducible LODO baseline for cross-domain ordinal DR grading on four public sources, characterises the cross-architecture variance, tests a pre-registered segmentation-auxiliary-channel ablation, and reports a per-threshold calibration diagnostic that independently confirms the prior-literature finding that post-hoc temperature scaling fails under DR cross-dataset shift. The pre-registered 2% QWK improvement threshold for the segmentation auxiliary was crossed on APTOS by the 4-ch soft variant (+2.19%) and on EyePACS by the same variant (+3.26%), just missed on DDR (+1.94%), and decisively *not* crossed on Messidor-2 (−4.57%); the cross-fold mean ΔQWK is +0.70% for 4-ch soft, −1.84% for 4-ch Tversky, and −2.16% for 4-ch morph. The class-balancing ablation shows the unbalanced baseline is uniformly +2.09% to

+5.67% better than the balanced baseline (mean ΔQWK +3.93%) — the dominant single ablation effect. The per-threshold calibration diagnostic (§4) shows both per-threshold and single-global temperature scaling *increase* held-out ECE (mean ΔECE +0.30 absolute), with the fitted global temperature saturating at $T = 5.0$ on every fold; this is an independent confirmation of the prior-literature finding (Poyrazer et al., 2026) reproduced here on a CORN ordinal regression head with the per-threshold decomposition. The honest reading is that the segmentation auxiliary channel is *not* a robust cross-domain improvement at the effect size pre-registered; the dominant negative ablation effect is the 3:1 class-balancing step; and the calibration diagnostic is the methodological contribution that survives both ablations.

9 Future Work

The analysis above is the complete set of empirical findings produced by this study. The following items are defined in the methodology or named in the results summary (`docs/results.md`) but are not yet executed or populated. Each item has a known compute budget and a known implementation surface; none requires methodological design work.

- **Bootstrap confidence intervals for all reported metrics.** *Completed 2026-08-12.* Per-sample predictions dumped to `saved/logs/per_sample_predictions/` for all five ConvNeXt-Tiny variants × four folds (20 inference runs, no retraining). 1000-resample paired bootstrap on the per-sample arrays yields the 95% confidence intervals for QWK, accuracy, and macro-F1 reported alongside the per-fold point estimates in ğ2–ğ4 and ğ6 of the results summary. JSON artifact: `saved/logs/bootstrap_cis.json`. Concrete observation: on APTOS the segmentation gain is statistically meaningful at $\alpha = 0.05$ (4-ch soft 0.862 [0.851, 0.872] vs 3-ch balanced 0.832 [0.822, 0.842], non-overlapping CIs); on EyePACS it is not (3-ch baseline 0.440 [0.432, 0.448] vs 4-ch soft 0.460 [0.453, 0.467], overlapping CIs).

- **Decision-boundary threshold tuning (ğ7 of the results summary).** *Completed 2026-08-12.* Post-hoc additive bias $b \in [-0.40, +0.40]$ on all four CORN sigmoid thresholds with honest 50/50 out-of-bias-sample CV (5 reps, stratified by class). Mean cross-variant ΔQWK is +0.025; every (variant, fold) cell except two sees non-negative ΔQWK. Largest single gain is 3-ch unbalanced × EyePACS (+0.0897, bias $b = -0.23$). No logits or weights are modified. Full per-(b) sweep curves in `docs/analysis/figures/decision_boundary/`.

- **Per-threshold calibration of the auxiliary-channel variants.** *Completed 2026-08-12.* Extended the per-threshold and global temperature scaling diagnostic to the 3-ch unbalanced, 4-ch soft, 4-ch Tversky, and 4-ch morph variants. Honest 50/50 5-rep CV (temperature fitted on train half, ECE on test half). JSON artifact: `saved/logs/calibration_all_variants.json`, markdown: `saved/logs/calibration_summary.md`. The global-temperature pathology ($T \to 5.0$ upper bound, mean ΔECE ≈ +0.10 to +0.18) is reproduced on every variant on every fold. The segmentation auxiliary does not rescue it.

- **Seed sensitivity for the high-priority variants (ğ9 of the results summary).** Repeat the 3-channel baseline (ConvNeXt-Tiny only) and the 4-channel soft-mask variant across a small pre-registered seed set (e.g., 3 seeds per fold). The seed set is to be pre-registered before any seed-sensitivity run is started. Estimated compute: 4 folds × 2 variants × 2 additional seeds = 16 runs at ~25 min each ≈ ~7 hours.

- **Single-source failure mode characterization (ğ10 of the results summary).** The qualitative failure-mode characterisation is now populated for the 3-channel ConvNeXt-Tiny baseline in this analysis (ğ5, Tables 3—6). The ğ10 table of the results summary is not yet populated for the full variant matrix.

- **Full backbone family.** The architecture comparison is across three backbones (ConvNeXt-Tiny, DeiT-3 Small/384, MaxViT-Tiny/384). ConvNeXt-Small (~50M parameters) and SwinV2-Base (192→384 transfer, ~88M parameters) are not evaluated. The runner, the loss function, and the LODO split are all in place; only the model yaml and the per-arch hyperparameter override (SwinV2's batch size 12) need to be configured. Estimated compute: ~5–8 hours of training across 8 folds.

- **Complete the auxiliary-channel ablation matrix.** The 4-channel Sobel variant, the 5-channel (soft + Sobel) variant, and the 5-channel (Tversky + Sobel) variant are not run in this study. The Sobel auxiliary pool is already pre-computed at preprocessing time; the 5-channel variants require a single training-time concatenation change. Estimated compute: ~12–14 runs × ~25 min/run on ConvNeXt-Tiny = ~5–6 hours.

- **Extend the source pool to the remaining four GDRBench sources.** The four sources used here are a strict subset of GDRBench's eight; the remaining four add cases for different acquisition protocols, different patient populations, and a different reference-grading scheme. The same pipeline applies without modification; the work is in the data acquisition and grading-scheme reconciliation, not in the modeling.

- **Investigate the temperature-saturation pathology.** The fitted global temperature saturates at the upper bound ($T = 5.0$) on every fold *and every variant tested* (the new all-variants diagnostic confirms it on 19/20 (variant, fold) cells), suggesting the optimizer is being aggressively pushed to extreme softening that does not transfer. Understanding why this happens — whether due to the validation-set sample size, the loss surface, or a bug in the temperature-scaling procedure — is a diagnostic follow-up before drawing strong conclusions about the post-hoc calibration failure mode.